E39: the third pre-registered confirmation of the pivot rules F, S and H

Run report and results, 2026-11-02. Research note `research/notes/index-calculus/RESEARCH_GLV_INVARIANT_FACTOR_BASES.md`: §9.48 (pre-registration, commit `16d17d2462`, pushed before any E39 code or run) and §9.49 (results). Code: `31f2a2999e` (the control hook `RuleCheck::bump_post` in `src/cryptanalysis/chain_pivot_rules.rs`, off by default) and `e1cd726577` (examples `e39_checks`, `e39_stage_a`, `e39_tables`; `identity.sh`, `run_e2e.sh`). Binaries built from a clean `git archive` export of `e1cd726577` (4,678 files checked blob by blob), pinned by sha256 in `experiments/46_pivot_rules_confirm3/binary_sha256.txt`. Records: `experiments/46_pivot_rules_confirm3/`. Host: the shared 4-vCPU Claude Code container. isolab-eks is not reachable from it and `conductor` is not installed. No isolation tier is earned: **operation counts only, no wall-time claim.**

This confirms a hypothesis that E36–E38 generated; it is not a new hypothesis. E36 and E37 measured F (each bucket's first-arrived row), S and H ahead of A and E25 at 0, 0.1 and 1 GAE on every row they ran, but in each a registered control failed and the kill voided the rank. E38 repaired the check on E37′s F control (`P_ext`) but its structure-preserving F control could never be aimed, so its kill fired at the pilot. E39 replaces that control with C_F^post and changes nothing else.

Outcome

- **No kill fired.** Identity 21/21 byte-identical; pilot P1 passed by every primary control, so no fallback was used; the main control gate held; Stage A and all 154 end-to-end records satisfy the identity clauses.
- **F, S and H are each ahead of A and of E25 at 0, 0.1 and 1 GAE on all 32 rows** (20 GLS + 12 E24), and stay ahead under σ1–σ3. With G2.191 and G2.192 holding, G2.201 holds: **F, S and H are ranked ahead of A and E25 across the band on these rows**: a measured engineering improvement of the `k = 4` solve's echelon (a constant). It has not passed the validation gate yet.
- Against B″, every rule is unresolved: B″ is ahead at 0 GAE and behind at 0.1 and 1. Rho is ahead of every arm on every row at every weight.

Controls

C_F^post: from the clean F run of the same call, `V` = the contested columns the left-looking normal forms read; `l*` = its lower median; after `l*`'s bucket updates (the pivot row's last use in the echelon) the pivot row's last-column entry (monomial 1, standard) is moved by one. `P_post`: pivot set and border targets unchanged, the first differing read is that row at `l*`, and a normal-form read of it changed. Before freezing, `V` ≠ ∅ was verified on all 160 of E38′s stored pilot calls (non-null `c_star`).

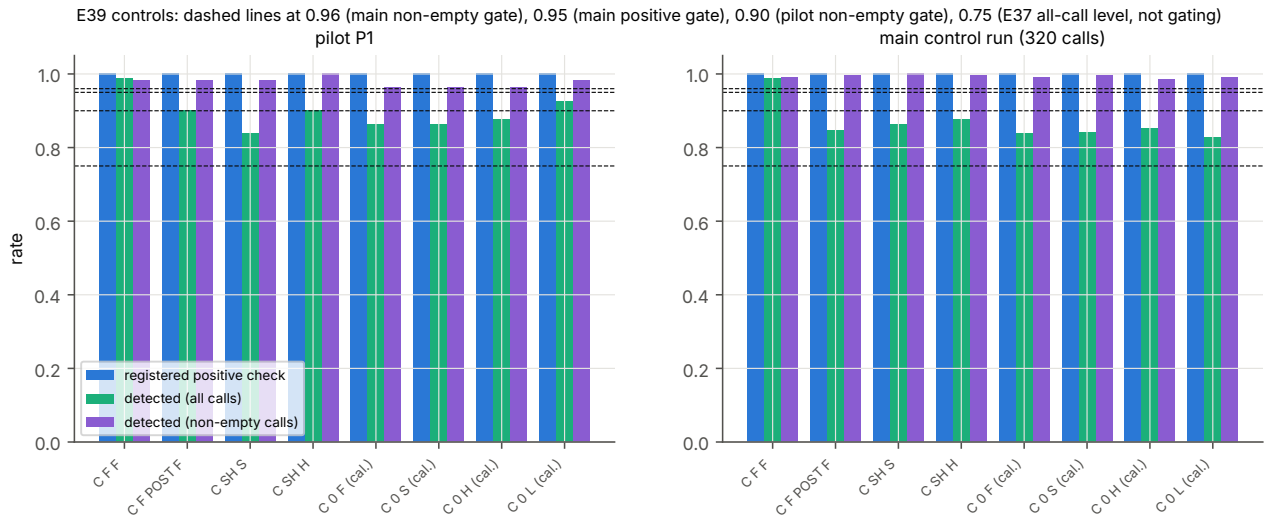


Figure 1: Pilot P1 (80 calls, 56 non-empty) and the main control run (320 calls, 194 non-empty). Gates: positive check at least 0.95 and non-empty detection at least 0.96 (from the pooled E37 and E38 C_SH and C_0 rates, lowest one-sided 95 % bound 0.9668). Source: `pilot.json`, `checks.json` via `summary.json`.

control	check	positive	detected (all)	non-empty detected	pilot non-empty
C_F (E37′s misfile)	`P_ext`	320/320	316/320	192/194	55/56
C_F^post (new)	`P_post`	320/320	271/320	193/194	55/56

C_SH (S)	strict	320/320	276/320	194/194	55/56
C_SH (H)	strict	320/320	280/320	193/194	56/56
C_0 (F, S, H, L), calibration	strict	320/320 each	265–273/320	191–193/194	54–55/56

C_F^post was aimed on 320/320 calls, with `|V|` 161–240 and `l*` in columns 255–302; pivot set and border targets were unchanged on all 320 and the first difference was the altered row's read on all 320. C_F changed the pivot set and the border targets on all 320 calls, and its strict check held on none, as in E38. The perturbed coefficient changed 194/194 non-empty sets under each of F, S, H, L; F's and L's digests differ from A's on 320/320. **Plainly:** C_F^post is mechanically close to the calibration C_0. It shows the detector sees a content corruption of a row F chose and the normal forms read; it does not test F's choice. C_F tests an F filing defect.

Stage A (10 curves × 1,000 fresh calls, salt `0xE39A0001`)

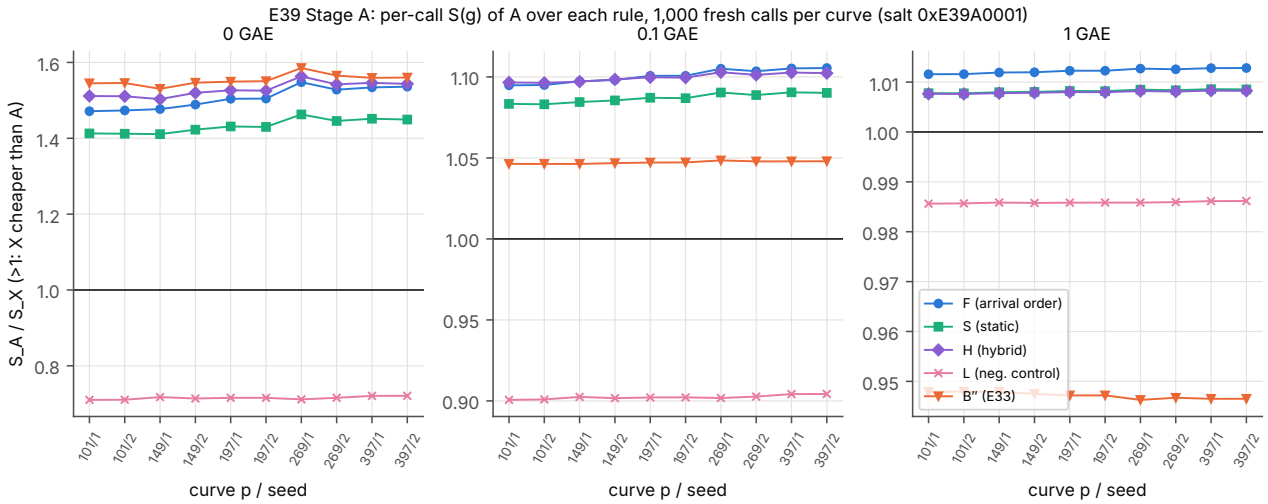


Figure 2: S_A/S_X per curve at 0, 0.1, 1 GAE (per-call means). Values from `tables.md` (computed by `e39_tables`).

Every solver returns the existing solver's outputs on every call, and the same calls fall back. Selection per call: F, L 0; S 511.9–512.7; H – S 333.95–334.00. C_A/C_X: F 1.471–1.548, S 1.411–1.463, H 1.503–1.563; C_A/C_L 0.710–0.721. S_A/S_X at 1 GAE: F 1.0116–1.0128, S 1.0077–1.0086, H 1.0076–1.0083; L 0.9856–0.9862.

End to end (32 rows, paired on the fresh stream `0xE39E0001`)

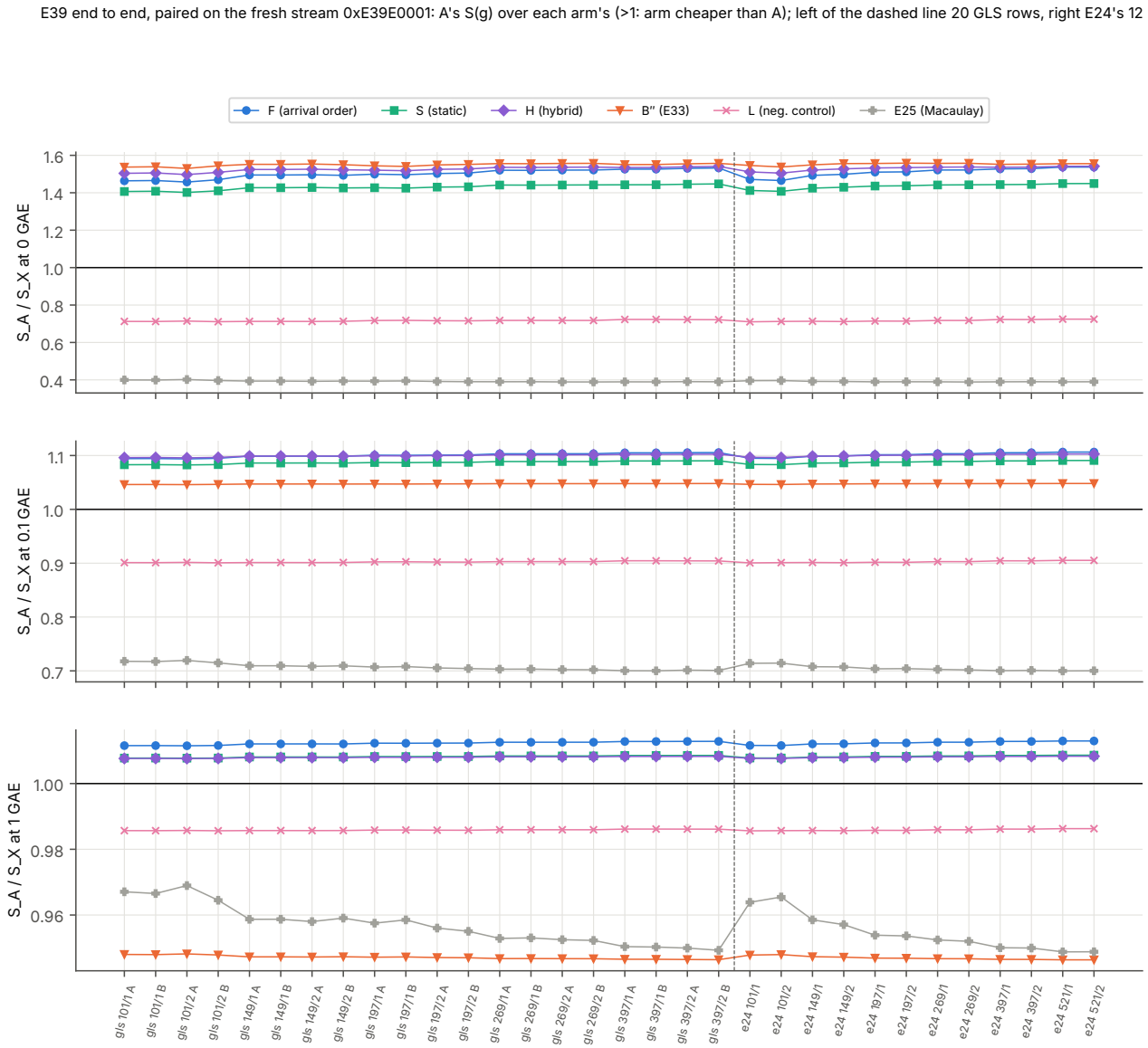


Figure 3: S_A/S_X per row at 0, 0.1, 1 GAE. Left of the dashed line E25′s 20 GLS rows, right E24′s 12 curves. Per-row values: `tables.md`.

X	vs	0 GAE	0.1 GAE	1 GAE	order, 32 rows
F	A	1.457–1.537	1.0937–1.1064	1.0115–1.0130	F ahead everywhere
F	E25	3.626–3.945	1.520–1.581	1.0439–1.0676	F ahead everywhere
F	B″	0.952–0.988	1.046–1.056	1.0668–1.0704	unresolved
S	A	1.402–1.449	1.0825–1.0908	1.0077–1.0086	S ahead everywhere
S	E25	3.488–3.719	1.505–1.559	1.0400–1.0631	S ahead everywhere
S	B″	0.913–0.932	1.035–1.041	1.0628–1.0658	unresolved
H	A	1.497–1.542	1.0957–1.1028	1.0076–1.0083	H ahead everywhere
H	E25	3.725–3.957	1.523–1.576	1.0399–1.0627	H ahead everywhere
H	B″	0.977–0.991	1.048–1.052	1.0627–1.0655	unresolved
B″	A	1.530–1.558	1.0459–1.0480	0.9464–0.9482	unresolved
L (control)	A	0.710–0.725	0.9006–0.9053	0.9856–0.9863	L behind everywhere

Values are S_{ref}/S_X (above 1, X is cheaper), $S(g) = \text{total_muls} + 97g(\text{entries} + \text{selection})$. Every F, S, H order holds under σ1–σ3. Under σ2 the control L moves ahead of A at 1 GAE on every row, an accounting effect on A that E36 and

E37 also saw and E39 predicted. Health: every log recovered, every walk correct, unsolved at most 0.05 %, fallbacks at most 1.30 % (E24 $p = 101$ seed 2: 1.13 %, ceiling 1.63 %), the same count for every chain arm on each row. A/A against E37's stream: $|\Delta S_A / S_F|$ at most 0.0111, 0.0010 and $8 \cdot 10^{-5}$.

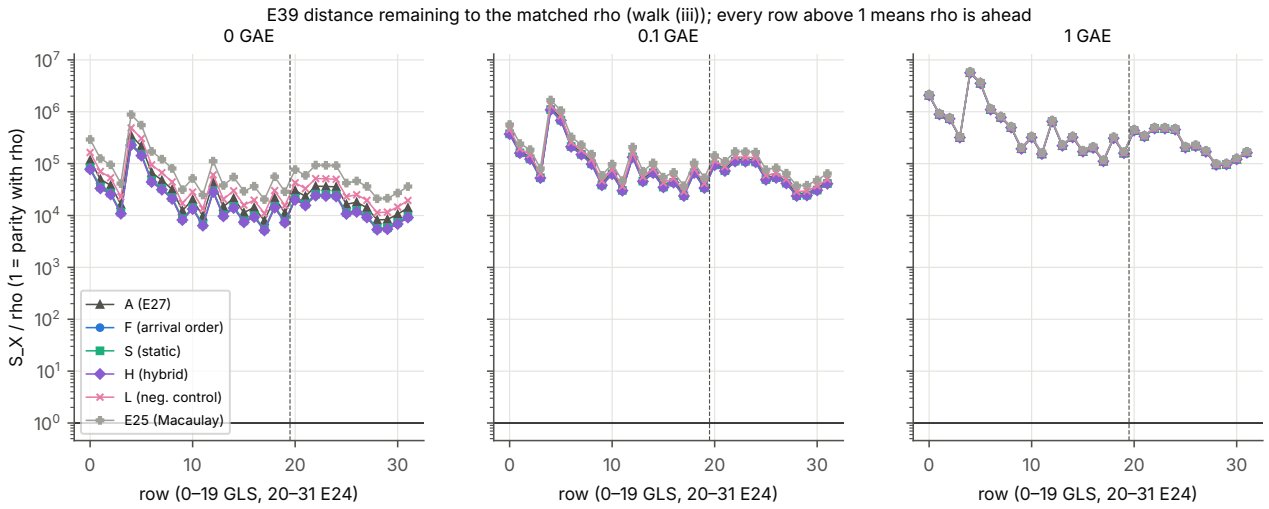


Figure 4: Distance remaining to the matched rho (walk (iii): E27's on GLS rows, E37's `walk_e24` on E24's, reproduced byte for byte by the E39 build), S_X/rho per row, log scale. Every row is above 1: rho is ahead.

Distance remaining for F: $5.18 \cdot 10^3$–$2.32 \cdot 10^5 \times$ rho at 0 GAE, and $9.40 \cdot 10^4$–$5.62 \cdot 10^6 \times$ at 1 GAE. The closest row, GLS $p = 397$ seed 1 arm B, has F at $5.18 \cdot 10^3 \times$ (A $7.91 \cdot 10^3 \times$) at 0 GAE and $1.11 \cdot 10^5 \times$ (A $1.12 \cdot 10^5 \times$) at 1 GAE. On E24's curves the closest row is $p = 397$ seed 1: $5.37 \cdot 10^3 \times$ and $9.40 \cdot 10^4 \times$. At 1 GAE the Macaulay entries are almost all of S, and no pivot rule changes them.

Verdicts

G2.190–G2.203 all hold. In particular G2.191 (identity) and G2.192 (controls) hold, so G2.201 holds: F, S and H are ranked ahead of A and E25 across the band on all 32 rows.

Limits

Local shared container: operation counts only; no isolation tier, no wall time. The σ2/σ3 end-to-end pricing is a Stage A per-call mean times the row's calls (the ten curves' mean for E24's rows), an estimate. The rank covers these 32 rows at $p \leq 521$ and the gain at 1 GAE is about 1 %. B″ remains unresolved against F, S and H. The validation gate (replay certificate, validator report, matched rho with the §5a band), the leaderboard and `benchmarks/` are left to the main session. PR 3307 merged while E39 ran (records up to Stage A); later commits are local.